RAJALAKSHMI ENGINEERING COLLEGE

Ajay Preethan K 251801010-2025-AIDS

A2

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|---|---|
| **Started on** | Thursday, 8 October 2026, 10:24 AM |
| **State** | Finished |
| **Completed on** | Thursday, 8 October 2026, 10:28 AM |
| **Time taken** | 4 mins 6 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    int a[n];
    int dp[n];

    for (int i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
        dp[i] = 1;
    }

    for (int i = 1; i < n; i++)
    {
        for (int j = 0; j < i; j++)
        {
            if (a[j] <= a[i])
            {
                if (dp[j] + 1 > dp[i])
                    dp[i] = dp[j] + 1;
            }
        }
    }

    int max = dp[0];

    for (int i = 1; i < n; i++)
    {
        if (dp[i] > max)
            max = dp[i];
    }

    printf("%d", max);

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 9<br>-1 3 4 5 2 2 2 2 3 | 6 | 6 | ✔ |
| ✔ | 7<br>1 2 2 4 5 7 6 | 6 | 6 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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